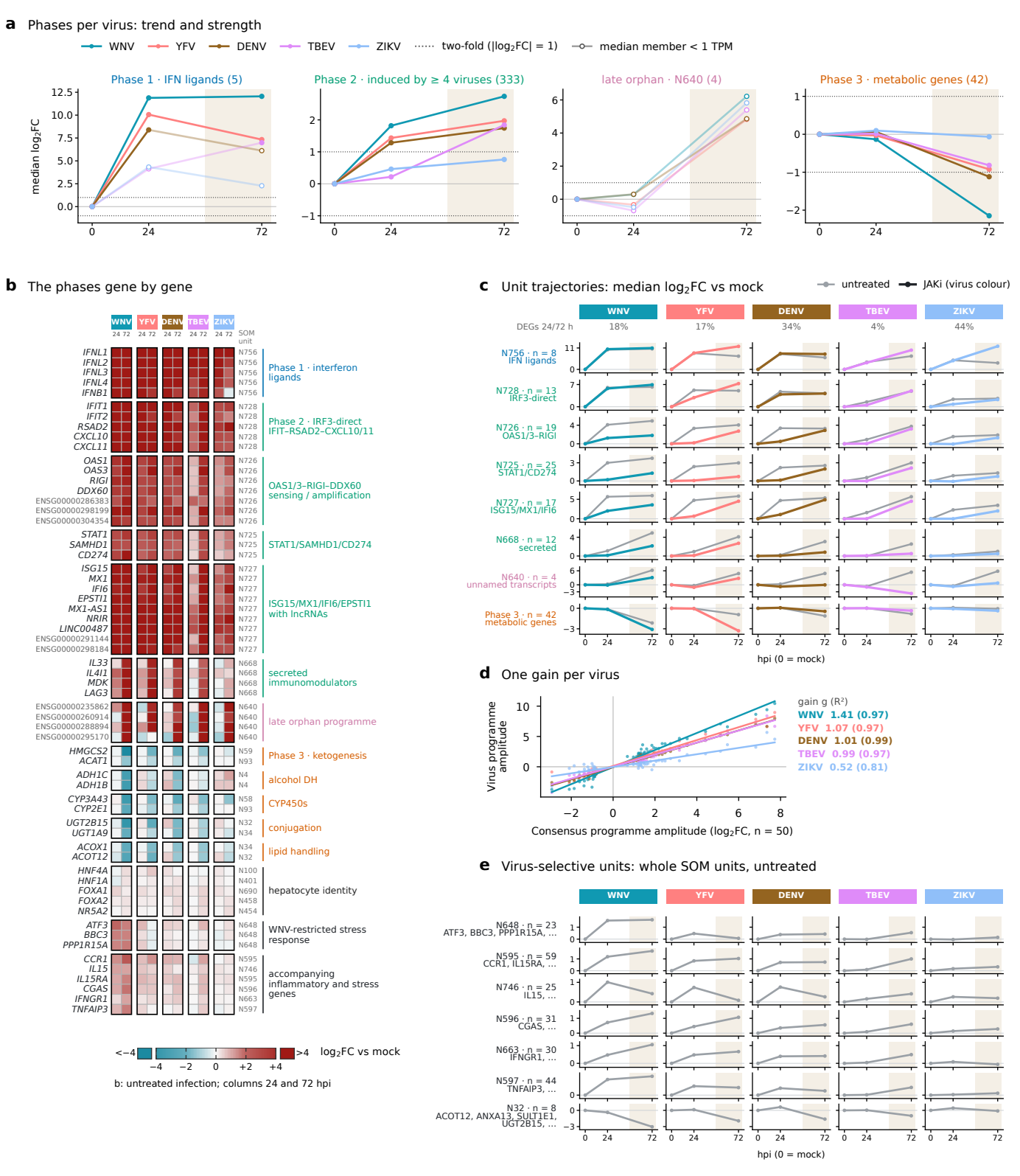

a
Phases per virus: trend and strength
WNV
YFV
DENV
TBEV
ZIKV
two-fold (|log2FC| = 1)
median member < 1 TPM
Phase 1 · IFN ligands (5)
Phase 2 · induced by ≥ 4 viruses (333)
late orphan · N640 (4)
Phase 3 · metabolic genes (42)
median log2FC
b
The phases gene by gene
SOM unit
IFNL1
IFNL2
IFNL3
IFNL4
IFNB1
Phase 1 · interferon ligands
IFIT1
IFIT2
RSAD2
CXCL10
CXCL11
Phase 2 · IRF3-direct IFIT–RSAD2–CXCL10/11
OAS1
OAS3
RIGI
DDX60
ENSG00000286383
ENSG00000298199
ENSG00000304354
OAS1/3–RIGI–DDX60 sensing / amplification
STAT1
SAMHD1
CD274
STAT1/SAMHD1/CD274
ISG15
MX1
IFI6
EPSTI1
MX1-AS1
NRIR
LINC00487
ENSG00000291144
ENSG00000298184
ISG15/MX1/IFI6/EPSTI1 with lncRNAs
IL33
IL4I1
MDK
LAG3
secreted immunomodulators
ENSG00000235862
ENSG00000260914
ENSG00000288894
ENSG00000295170
late orphan programme
HMGCS2
ACAT1
Phase 3 · ketogenesis
ADH1C
ADH1B
alcohol DH
CYP3A43
CYP2E1
CYP450s
UGT2B15
UGT1A9
conjugation
ACOX1
ACOT12
lipid handling
HNF4A
HNF1A
FOXA1
FOXA2
NR5A2
hepatocyte identity
ATF3
BBC3
PPP1R15A
WNV-restricted stress response
CCR1
IL15
IL15RA
CGAS
IFNGR1
TNFAIP3
accompanying inflammatory and stress genes
<−4
>4
log2FC vs mock
b: untreated infection; columns 24 and 72 hpi
c
Unit trajectories: median log2FC vs mock
untreated
JAKi (virus colour)
DEGs 24/72 h
18%
17%
34%
4%
44%
N756 · n = 8
IFN ligands
N728 · n = 13
IRF3-direct
N726 · n = 19
OAS1/3–RIGI
N725 · n = 25
STAT1/CD274
N727 · n = 17
ISG15/MX1/IFI6
N668 · n = 12
secreted
N640 · n = 4
unnamed transcripts
Phase 3 · n = 42
metabolic genes
hpi (0 = mock)
d
One gain per virus
Virus programme amplitude
Consensus programme amplitude (log2FC, n = 50)
gain g (R²)
WNV 1.41 (0.97)
YFV 1.07 (0.97)
DENV 1.01 (0.99)
TBEV 0.99 (0.97)
ZIKV 0.52 (0.81)
e
Virus-selective units: whole SOM units, untreated
N648 · n = 23
ATF3, BBC3, PPP1R15A, …
N595 · n = 59
CCR1, IL15RA, …
N746 · n = 25
IL15, …
N596 · n = 31
CGAS, …
N663 · n = 30
IFNGR1, …
N597 · n = 44
TNFAIP3, …
N32 · n = 8
ACOT12, ANXA13, SULT1E1, UGT2B15, …
hpi (0 = mock)